

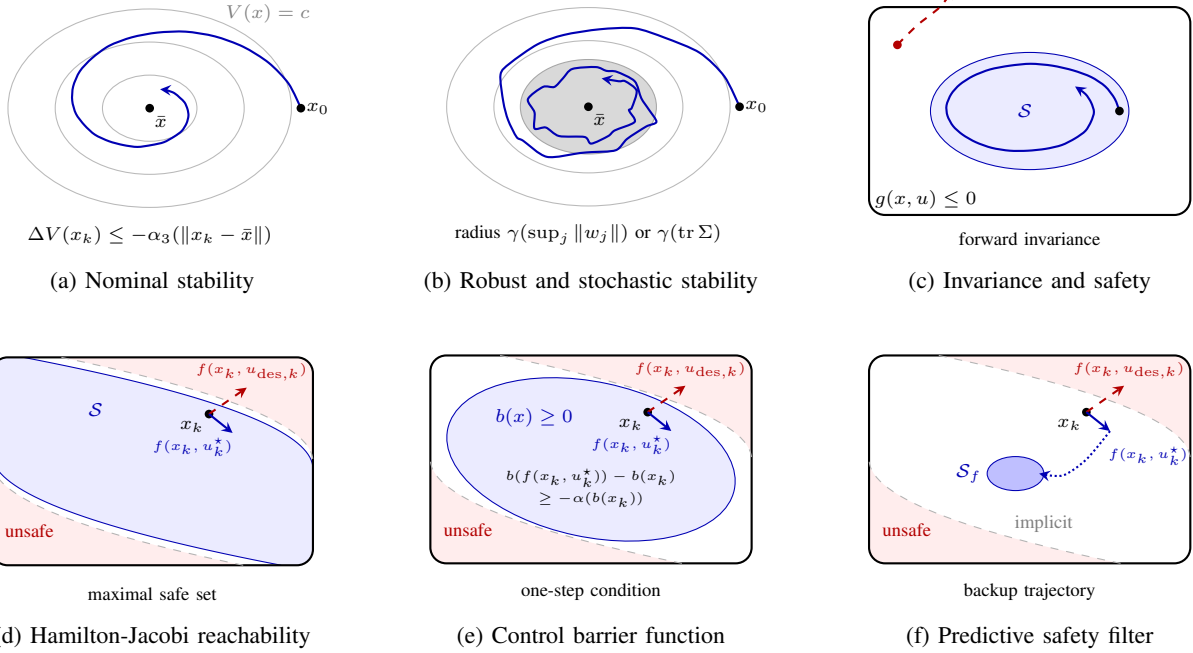


Fig. 1: Illustration of the stability and safety notions of this section. (a) Nominal stability: a Lyapunov function  $V$  decreases along closed-loop trajectories (level sets in gray) and the state converges to the equilibrium  $\bar{x}$ . (b) Robust and stochastic stability: under disturbances  $w$ , the state converges to a neighbourhood of  $\bar{x}$  (shaded), whose size depends on the disturbance bound or the noise covariance. (c) Safety: constraint satisfaction at one time instant does not imply safety (red dashed); safety is guaranteed by an invariant set  $\mathcal{S}$  contained in the constraints. (d)–(f) Safety filters: the desired input  $u_{\text{des},k}$  would leave the safe set and is modified to  $u_k^*$ ; the unsafe states are shaded red (edge dashed). HJ reachability computes the maximal safe set  $\mathcal{S}$  offline, the largest invariant set within the constraints. A CBF certifies invariance of  $\{b(x) \geq 0\}$  by the one-step condition on  $b(x_{k+1})$ . A PSF does not compute the safe set but certifies safety by a backup trajectory (dotted) into a smaller terminal set  $\mathcal{S}_f$ .